ESP32 – RESEARCH

An **ESP32** is a low-cost, low-power System-on-Chip (SoC) microcontroller created by Espressif Systems that features built-in Wi-Fi and Bluetooth capabilities.

- **Wireless:** Integrated **2.4 GHz Wi-Fi** (802.11 b/g/n) and **Bluetooth 4.2 / BLE** (Bluetooth Low Energy) with an onboard antenna.
- **Memory:** Typically includes **520 KB of SRAM** and external flash memory (often 4 MB to 16 MB).
- **Input/Output (Peripherals):** Dozens of **GPIO** (General Purpose Input/Output) pins, Analog-to-Digital Converters (ADCs), Digital-to-Analog Converters (DACs), touch-sensing pins, and communication interfaces like I2C, SPI, and UART.

How It Works

- **Writing Code:** You write a program (firmware) on your computer using tools like the **Arduino IDE, MicroPython,** or the **ESP-IDF** (Espressif IoT Development Framework).
- **Flashing the Chip:** You connect the ESP32 board to your computer via a USB cable and upload (flash) your compiled code directly into its onboard flash memory.
- **Interacting with Hardware:** Once powered on, the ESP32 executes your code loop-by-loop. It reads data from physical sensors (like temperature or motion detectors) or controls actuators (like motors and LED lights) via its GPIO pins.
- **Connecting Wirelessly:** Using its built-in radio, the ESP32 can connect to a local Wi-Fi router to send sensor data to the cloud, host a local web server, or talk directly to other ESP32 boards using peer-to-peer protocols like **ESP-NOW**.

ESP32 DevKit V1:

The **ESP32 DevKit V1** pin configuration provides multi-functional General Purpose Input/Output (GPIO) pins, power rails, and communication interfaces operating at 3.3V logic (note that ESP32 pins are **not** 5V tolerant).

Analog Input (ADC)

The ESP32 has **18 Analog-to-Digital Converter (ADC) channels** with up to **12-bit resolution** (configurable from 9 to 12 bits, giving values from 0 to 4095).

- **Safe Input Voltage Range: 0V to 3.3V**. Applying anything higher will permanently damage the chip.
- **ADC1 (8 Channels - Safe to use anytime):**
  - GPIO 32, 33, 34, 35, 36, 37, 38, 39
- **ADC2 (10 Channels - Restricted):**
  - GPIO 0, 2, 4, 12, 13, 14, 15, 25, 26, 27
  - **Critical Note: ADC2 cannot be used when Wi-Fi is active.** If your project uses Wi-Fi, you must stick strictly to ADC1 (GPIOs 32–39).
- **Input-Only Pins:** GPIOs 34, 35, 36, and 39 are hardware-bound input-only pins, making them the absolute best choice for analog sensors because you cannot accidentally set them as outputs

Supply Voltage:

ESP32 DevKit V1 using three primary methods:

- **Micro-USB Port (5V):** The easiest method. You plug it into a computer or a 5V USB wall adapter. The onboard voltage regulator drops this down to the required 3.3V.
- **VIN Pin (5V to 12V):** If you are using an external power source (like a battery pack or DC adapter), connect the positive lead to **VIN** and the ground to **GND**. The onboard regulator handles the conversion. A stable **5V to 9V** input is ideal; going up to 12V is acceptable but will make the onboard regulator run hot.
- **3V3 Pin (Strictly 3.3V):** You can bypass the onboard regulator completely by feeding exactly **3.3V** directly into the 3V3 pin. *Warning: This pin has no reverse-voltage or over-voltage protection. If you accidentally feed 5V here, you will instantly fry the board.*

WiFi Capabilities:

Yes, the ESP32 DevKit V1 has built-in 2.4 GHz WiFi .
along with Bluetooth v4.2 .

This native internet connectivity makes it a popular choice for **future mobile alerts**. You can easily program the board to trigger and send real-time push notifications, SMS alerts, or emails directly to your smartphone using free IoT platforms and webhooks like **Blynk, IFTTT, Twilio, or Firebase**.

I will be using firebase as alert system later but the proble Is that If I try to make the esp32 by myself then there will be a shortage of time which can ruine the whole model so I have to work accordingly to make my model on time

And of quality.

Price in India:

standalone **ESP32 DevKit V1** is very affordable and typically costs between **₹300 and ₹600**
, depending on the exact pin configuration (30-pin vs. 38-pin variants) and the integrated USB-to-UART controller chip .

- **₹308** for the Type-C USB CH340C variant at Autables
- **₹365** for the Doit DevKit V1 module at techiesms --- My choice for now
- **₹373** for a standard 30-pin board at KTRON India
